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Enum ActivationDataType

Namespace: Uralstech.UAI.LiteRTLM

```
public enum ActivationDataType
```

Fields

`Float16 = 1`

Use float16 as the activation data type.

`Float32 = 0`

Use float32 as the activation data type.

`Int16 = 2`

Use int16 as the activation data type.

`Int8 = 3`

Use int8 as the activation data type.

Class BackendNames

Namespace: Uralstech.UAI.LiteRTLM

```
public static class BackendNames
```

Inheritance

object ← BackendNames

Fields

CPU

```
public const string CPU = "cpu"
```

Field Value

string

CPUArtisan

```
public const string CPUArtisan = "cpu_artisan"
```

Field Value

string

GPU

```
public const string GPU = "gpu"
```

Field Value

string

GPUArtisan

```
public const string GPUArtisan = "gpu_artisan"
```

Field Value

string

GoogleTensorArtisan

```
public const string GoogleTensorArtisan = "google_tensor_artisan"
```

Field Value

string

NPU

```
public const string NPU = "npu"
```

Field Value

string

Class BenchmarkInfo

Namespace: Uralstech.UAI.LiteRTLM

```
public sealed class BenchmarkInfo : LiteRTLMNativeHandle
```

Inheritance

object ← LiteRTLMNativeHandle ← BenchmarkInfo

Inherited Members

LiteRTLMNativeHandle.Dispose()

Methods

GetDecodeTurns()

Returns the benchmark information for each decode turn.

```
public BenchmarkInfo.Turn[] GetDecodeTurns()
```

Returns

Turn[]

GetPrefillTurns()

Returns the benchmark information for each prefill turn.

```
public BenchmarkInfo.Turn[] GetPrefillTurns()
```

Returns

Turn[]

GetTimeToFirstToken()

Returns the time to the first token in seconds.

```
public double GetTimeToFirstToken()
```

Returns

double

The time to the first token in seconds.

Remarks

Note that the time to first token does not include initialization time. It is the sum of the prefill time for the first turn and the time spent decoding the first token.

GetTotalInitTime()

Returns the total initialization time in seconds.

```
public double GetTotalInitTime()
```

Returns

double

The total initialization time in seconds.

ReleaseUnmanagedResources()

```
protected override void ReleaseUnmanagedResources()
```

Struct BenchmarkInfo.Turn

Namespace: Uralstech.UAI.LiteRTLM

Benchmarks for a prefill or decode turn.

```
public readonly struct BenchmarkInfo.Turn
```

Constructors

Turn(int, double)

```
public Turn(int tokenCount, double tokensPerSecond)
```

Parameters

`tokenCount` int

`tokensPerSecond` double

Fields

TokenCount

The prefill/decode token count for this turn.

```
public readonly int TokenCount
```

Field Value

int

TokensPerSecond

The prefill/decode tokens per second for this turn.

```
public readonly double TokensPerSecond
```

Field Value

double

Class Conversation

Namespace: Uralstech.UAI.LiteRTLM

```
public sealed class Conversation : LiteRTLMNativeHandle
```

Inheritance

object ← LiteRTLMNativeHandle ← Conversation

Inherited Members

LiteRTLMNativeHandle.Dispose()

Constructors

Conversation(Engine, ConversationConfig?)

Creates a managed wrapper around a LiteRT LM conversation. The caller is responsible for disposing the wrapper using Dispose().

```
public Conversation(Engine engine, ConversationConfig? config = null)
```

Parameters

`engine` Engine

The engine to create the conversation from.

`config` ConversationConfig

The conversation configuration to use. If null, the default configuration is used.

Exceptions

InvalidOperationException

Thrown if the native object could not be created.

Methods

CancelProcess()

Cancels the ongoing inference process for asynchronous inference.

```
public void CancelProcess()
```

Clone()

Clones a LiteRT LM conversation, duplicating its prefilled state. The caller is responsible for disposing the cloned wrapper using Dispose().

```
public Conversation? Clone()
```

Returns

Conversation

The cloned conversation wrapper, or null on failure.

GetBenchmarkInfo()

Retrieves benchmark information from the conversation. The caller is responsible for disposing the returned wrapper.

```
public BenchmarkInfo? GetBenchmarkInfo()
```

Returns

BenchmarkInfo

The benchmark information wrapper, or null on failure.

GetTokenCount()

Gets the number of tokens in the conversation KV cache (prefill + decode).

```
public int GetTokenCount()
```

Returns

int

The number of tokens, or a negative value on failure.

ReleaseUnmanagedResources()

```
protected override void ReleaseUnmanagedResources()
```

RenderMessageToString(string)

Renders the message into a string according to the template.

```
public string? RenderMessageToString(string messageJson)
```

Parameters

`messageJson` string

A JSON string representing the message to render.

Returns

string

The rendered string, or null on failure.

Remarks

This function does not need to be called for actual message sending, as SendMessage(string, string?, ConversationOptionalArgs?) and SendMessageStream(StreamCallback, string, string?, ConversationOptionalArgs?) render messages internally.

RenderPrefaceToString()

Renders the preface into a string according to the template.

```
public string? RenderPrefaceToString()
```

Returns

string

The rendered string, or null on failure.

SendMessage(string, string?, ConversationOptionalArgs?)

Sends a message to the conversation and returns the response. This is a blocking call.

```
public JsonResponse? SendMessage(string messageJson, string? extraContext = null,
ConversationOptionalArgs? optionalArgs = null)
```

Parameters

`messageJson` string

A JSON string representing the message to send.

`extraContext` string

A JSON string representing the extra context to use.

`optionalArgs` ConversationOptionalArgs

The optional arguments to use.

Returns

JsonResponse

A JSON response wrapper, or null on failure. The caller is responsible for disposing the returned wrapper.

SendMessageStream(StreamCallback, string, string?, ConversationOptionalArgs?)

Sends a message to the conversation and streams the response through a callback. This is a non-blocking call that invokes the callback from a background thread for each chunk.

```
public int SendMessageStream(StreamCallback callback, string messageJson, string?
extraContext = null, ConversationOptionalArgs? optionalArgs = null)
```

Parameters

`callback` [StreamCallback]

The callback function ([StreamCallback]) that receives response chunks.

`messageJson` string

A JSON string representing the message to send.

`extraContext` string

A JSON string representing the extra context to use.

`optionalArgs` [ConversationOptionalArgs]

The optional arguments to use.

Returns

int

0 on success, non-zero on failure to start the stream.

Class ConversationConfig

Namespace: [Uralstech](#).[UAI](#).[LiteRTLM](#)

```
public sealed class ConversationConfig : LiteRTLMNativeHandle
```

Inheritance

object ← [LiteRTLMNativeHandle](#) ← ConversationConfig

Inherited Members

[LiteRTLMNativeHandle.Dispose()](#)

Constructors

ConversationConfig()

Creates a managed wrapper around a LiteRT LM conversation configuration. The caller is responsible for disposing the wrapper using [Dispose()](#).

```
public ConversationConfig()
```

Exceptions

InvalidOperationException

Thrown if the native object could not be created.

Methods

ReleaseUnmanagedResources()

```
protected override void ReleaseUnmanagedResources()
```

SetEnableConstrainedDecoding(bool)

Sets whether to enable constrained decoding for this conversation configuration.

```
public void SetEnableConstrainedDecoding(bool enableConstrainedDecoding)
```

Parameters

`enableConstrainedDecoding` bool

Whether to enable constrained decoding.

SetExtraContext(string)

Sets the extra context for the conversation preface.

```
public void SetExtraContext(string extraContextJson)
```

Parameters

`extraContextJson` string

A JSON string representing the extra context object.

SetFilterChannelContentFromKvCache(bool)

Sets whether to filter channel content from the KV cache.

```
public void SetFilterChannelContentFromKvCache(bool filterChannelContentFromKvCache)
```

Parameters

`filterChannelContentFromKvCache` bool

Whether to filter channel content.

SetMessages(string)

Sets the initial messages for this conversation configuration.

```
public void SetMessages(string messagesJson)
```

Parameters

`messagesJson` string

The initial messages in JSON array format.

SetSessionConfig(SessionConfig)

Sets the session config for this conversation configuration.

```
public void SetSessionConfig(SessionConfig sessionConfig)
```

Parameters

`sessionConfig` [SessionConfig]

The session config to use.

SetStreamToolCalls(bool, string)

Sets whether to stream tool call tokens.

```
public void SetStreamToolCalls(bool streamToolCalls, string channelName)
```

Parameters

`streamToolCalls` bool

Whether to stream tool call tokens.

`channelName` string

The channel name to use for tool call tokens.

SetSystemMessage(string)

Sets the system message for this conversation configuration.

```
public void SetSystemMessage(string systemMessageJson)
```

Parameters

`systemMessageJson` string

The system message in JSON format.

SetThinkingConfig(ThinkingConfig?)

Sets the thinking config for this conversation config.

```
public void SetThinkingConfig(ThinkingConfig? thinkingConfig)
```

Parameters

`thinkingConfig` [ThinkingConfig]

The thinking config to set. If [null], clears any previously set thinking config.

SetTools(string)

Sets the tools for this conversation configuration.

```
public void SetTools(string toolsJson)
```

Parameters

`toolsJson` string

The tools description in JSON array format.

Class ConversationOptionalArgs

Namespace: Uralstech.UAI.LiteRTLM

```
public sealed class ConversationOptionalArgs : LiteRTLMNativeHandle
```

Inheritance

object ← LiteRTLMNativeHandle ← ConversationOptionalArgs

Inherited Members

LiteRTLMNativeHandle.Dispose()

Constructors

ConversationOptionalArgs()

Creates a managed wrapper around the optional arguments for conversation APIs. The caller is responsible for disposing the wrapper using Dispose().

```
public ConversationOptionalArgs()
```

Exceptions

InvalidOperationException

Thrown if the native object could not be created.

Methods

ReleaseUnmanagedResources()

```
protected override void ReleaseUnmanagedResources()
```

SetMaxOutputTokens(int)

Sets the maximum number of output tokens for the conversation optional arguments.

```
public void SetMaxOutputTokens(int maxOutputTokens)
```

Parameters

`maxOutputTokens` int

The maximum number of output tokens.

SetRepetitionPenaltyConfig(RepetitionPenaltyConfig?)

Sets the repetition penalty configuration for the conversation optional args.

```
public void SetRepetitionPenaltyConfig(RepetitionPenaltyConfig? repetitionPenaltyConfig)
```

Parameters

`repetitionPenaltyConfig` RepetitionPenaltyConfig

The repetition penalty config to set. If null, clears any previously set repetition penalty config.

SetThinkingConfig(ThinkingConfig?)

Sets the thinking config for the conversation optional args.

```
public void SetThinkingConfig(ThinkingConfig? thinkingConfig)
```

Parameters

`thinkingConfig` ThinkingConfig

The thinking config to set. If null, clears any previously set thinking config.

SetVisualTokenBudget(int)

Sets the visual token budget for the conversation optional arguments.

```
public void SetVisualTokenBudget(int visualTokenBudget)
```

Parameters

`visualTokenBudget` int

The visual token budget.

Class DetokenizeResult

Namespace: Uralstech.UAI.LiteRTLM

```
public sealed class DetokenizeResult : LiteRTLMNativeHandle
```

Inheritance

object ← LiteRTLMNativeHandle ← DetokenizeResult

Inherited Members

LiteRTLMNativeHandle.Dispose()

Methods

GetString()

Returns the string.

```
public string GetString()
```

Returns

string

The detokenized string.

ReleaseUnmanagedResources()

```
protected override void ReleaseUnmanagedResources()
```

Class Engine

Namespace: Uralstech.UAI.LiteRTLM

```
public sealed class Engine : LiteRTLMNativeHandle
```

Inheritance

object ← LiteRTLMNativeHandle ← Engine

Inherited Members

LiteRTLMNativeHandle.Dispose()

Constructors

Engine(EngineSettings)

Creates a managed wrapper around a LiteRT LM engine from the given settings. The caller is responsible for disposing the wrapper using Dispose().

```
public Engine(EngineSettings settings)
```

Parameters

`settings` EngineSettings

The engine settings.

Exceptions

InvalidOperationException

Thrown if the native object could not be created.

Methods

CreateSession(SessionConfig?)

Creates a managed wrapper around a LiteRT LM session. The caller is responsible for disposing the wrapper using Dispose().

```
public Session? CreateSession(SessionConfig? config = null)
```

Parameters

`config` SessionConfig

The session configuration to use. If null, the default session configuration is used.

Returns

Session

The created session wrapper, or null on failure.

Detokenize(int[])

Detokenizes token ids using the engine's tokenizer.

```
public DetokenizeResult? Detokenize(int[] tokens)
```

Parameters

`tokens` int[]

An array of token ids to detokenize.

Returns

DetokenizeResult

The detokenize result wrapper, or null on failure. The caller is responsible for disposing the returned wrapper.

GetStartToken()

Returns the configured start token (BOS), if any.

```
public TokenUnion? GetStartToken()
```

Returns

TokenUnion

The start token wrapper, or null if none is configured. The caller is responsible for disposing the returned wrapper.

GetStopTokens()

Returns the configured stop tokens (EOS).

```
public TokenUnions? GetStopTokens()
```

Returns

TokenUnions

The stop tokens wrapper, or null if none are configured. The caller is responsible for disposing the returned wrapper.

ReleaseUnmanagedResources()

```
protected override void ReleaseUnmanagedResources()
```

Tokenize(string)

Tokenizes text using the engine's tokenizer.

```
public TokenizeResult? Tokenize(string text)
```

Parameters

`text` string

The UTF-8 string to tokenize.

Returns

TokenizeResult

The tokenize result wrapper, or null on failure. The caller is responsible for disposing the returned wrapper.

Class EngineSettings

Namespace: Uralstech.UAI.LiteRTLM

```
public sealed class EngineSettings : LiteRTLMNativeHandle
```

Inheritance

object ← LiteRTLMNativeHandle ← EngineSettings

Inherited Members

LiteRTLMNativeHandle.Dispose()

Constructors

EngineSettings(int, string, string?, string?)

Creates managed engine settings from a raw file descriptor. The engine takes ownership of the file descriptor and closes it when done. The caller is responsible for disposing the wrapper using Dispose().

```
public EngineSettings(int fd, string backend, string? visionBackend = null, string?
audioBackend = null)
```

Parameters

`fd` int

The file descriptor of the model.

`backend` string

The backend to use (for example, "cpu" or "gpu").

`visionBackend` string

The vision backend to use, or null if not set.

`audioBackend` string

The audio backend to use, or null if not set.

Remarks

See BackendNames for valid backend strings.

Exceptions

InvalidOperationException

Thrown if the native object could not be created.

EngineSettings(string, string, string?, string?)

Creates managed engine settings from a model path. The caller is responsible for disposing the wrapper using Dispose().

```
public EngineSettings(string modelPath, string backend, string? visionBackend = null,
string? audioBackend = null)
```

Parameters

`modelPath` string

The path to the model file.

`backend` string

The backend to use (for example, "cpu" or "gpu").

`visionBackend` string

The vision backend to use, or null if not set.

`audioBackend` string

The audio backend to use, or null if not set.

Remarks

See BackendNames for valid backend strings.

Exceptions

InvalidOperationException

Thrown if the native object could not be created.

Methods

EnableBenchmark()

Enables benchmarking for the engine.

```
public void EnableBenchmark()
```

ReleaseUnmanagedResources()

```
protected override void ReleaseUnmanagedResources()
```

SetActivationDataType(ActivationDataType)

Sets the activation data type.

```
public void SetActivationDataType(ActivationDataType activationDataType)
```

Parameters

`activationDataType` ActivationDataType

The activation data type.

SetAudioLoraRank(int)

Sets the Audio LoRA rank for the engine.

```
public void SetAudioLoraRank(int loraRank)
```

Parameters

`loraRank` int

The Audio LoRA rank.

SetAudioNumThreads(int)

Sets the number of threads for the audio CPU backend.

```
public void SetAudioNumThreads(int numThreads)
```

Parameters

`numThreads` int

The number of threads.

SetCacheDir(string)

Sets the cache directory for the engine.

```
public void SetCacheDir(string cacheDir)
```

Parameters

`cacheDir` string

The cache directory.

SetEnableSpeculativeDecoding(bool)

Sets whether to enable speculative decoding.

```
public void SetEnableSpeculativeDecoding(bool enableSpeculativeDecoding)
```

Parameters

`enableSpeculativeDecoding` bool

Whether to enable speculative decoding.

SetLitertDispatchLibDir(string)

Sets the LiteRT dispatch library directory for the NPU backend.

```
public void SetLitertDispatchLibDir(string libDir)
```

Parameters

`libDir` string

The dispatch library directory.

SetLoraRank(int)

Sets the LoRA rank for the engine.

```
public void SetLoraRank(int loraRank)
```

Parameters

`loraRank` int

The LoRA rank.

SetMaxNumImages(int)

Sets the maximum number of images for the engine. This is only used for the legacy implementation of the engine.

```
public void SetMaxNumImages(int maxNumImages)
```

Parameters

`maxNumImages` int

The maximum number of images.

SetMaxNumTokens(int)

Sets the maximum number of tokens for the engine.

```
public void SetMaxNumTokens(int maxNumTokens)
```

Parameters

`maxNumTokens` int

The maximum number of tokens.

SetNumDecodeTokens(int)

Sets the number of decode tokens for benchmarking.

```
public void SetNumDecodeTokens(int numDecodeTokens)
```

Parameters

`numDecodeTokens` int

The number of decode tokens.

SetNumPrefillTokens(int)

Sets the number of prefill tokens for benchmarking.

```
public void SetNumPrefillTokens(int numPrefillTokens)
```

Parameters

`numPrefillTokens` int

The number of prefill tokens.

SetNumThreads(int)

Sets the number of threads for the CPU backend.

```
public void SetNumThreads(int numThreads)
```

Parameters

`numThreads` int

The number of threads.

SetParallelFileSectionLoading(bool)

Sets whether the engine should load different sections of the litertlm file in parallel. Defaults to true.

```
public void SetParallelFileSectionLoading(bool parallelFileSectionLoading)
```

Parameters

`parallelFileSectionLoading` bool

Whether to load in parallel.

SetPrefillChunkSize(int)

Sets the prefill chunk size for the engine. Only applicable for the CPU backend with dynamic models.

```
public void SetPrefillChunkSize(int prefillChunkSize)
```

Parameters

`prefillChunkSize` int

The prefill chunk size.

SetSupportedAudioLoraRanks(int[])

Sets the supported Audio LoRA ranks for the engine.

```
public int SetSupportedAudioLoraRanks(int[] loraRanks)
```

Parameters

`loraRanks` int[]

An array of supported Audio LoRA ranks.

Returns

int

0 on success, non-zero on failure.

SetSupportedLoraRanks(int[])

Sets the supported LoRA ranks for the engine.

```
public int SetSupportedLoraRanks(int[] loraRanks)
```

Parameters

`loraRanks` int[]

An array of supported LoRA ranks.

Returns

int

0 on success, non-zero on failure.

Class InputData

Namespace: Uralstech.UAI.LiteRTLM

```
public sealed class InputData : LiteRTLMNativeHandle
```

Inheritance

object ← LiteRTLMNativeHandle ← InputData

Inherited Members

LiteRTLMNativeHandle.Dispose()

Constructors

InputData(string)

Creates a managed wrapper around text input data. The caller is responsible for disposing the wrapper using Dispose().

```
public InputData(string data)
```

Parameters

`data` string

The text data.

Exceptions

ArgumentException

Thrown if `data` is null or empty.

InvalidOperationException

Thrown if the native object could not be created.

InputData(InputDataType, nint, nuint)

Creates a managed wrapper around LiteRT LM input data. The caller is responsible for disposing the wrapper using Dispose().

```
public InputData(InputDataType dataType, nint nativeData, nuint size)
```

Parameters

`dataType` InputDataType

The type of the input data.

`nativeData` nint

The data buffer. The data is copied internally.

`size` nuint

The size of the data in bytes.

Exceptions

ArgumentException

Thrown if `nativeData` or `size` is invalid.

InvalidOperationException

Thrown if the native object could not be created.

InputData(InputDataType, ReadOnlySpan<byte>)

Creates a managed wrapper around binary input data. The caller is responsible for disposing the wrapper using Dispose().

```
public InputData(InputDataType dataType, ReadOnlySpan<byte> data)
```

Parameters

`dataType` InputDataType

The type of the input data.

`data` ReadOnlySpan<byte>

The data buffer.

Exceptions

ArgumentException

Thrown if `data` is empty.

InvalidOperationException

Thrown if the native object could not be created.

Methods

ReleaseUnmanagedResources()

```
protected override void ReleaseUnmanagedResources()
```

Enum InputDataType

Namespace: Uralstech.UAI.LiteRTLM

Represents the type of input data.

```
public enum InputDataType
```

Fields

`Audio = 3`

`AudioEnd = 4`

`Image = 1`

`ImageEnd = 2`

`Text = 0`

A UTF-8 string.

Class JsonResponse

Namespace: Uralstech.UAI.LiteRTLM

```
public sealed class JsonResponse : LiteRTLMNativeHandle
```

Inheritance

object ← LiteRTLMNativeHandle ← JsonResponse

Inherited Members

LiteRTLMNativeHandle.Dispose()

Methods

GetString()

Returns the JSON response string.

```
public string GetString()
```

Returns

string

The response JSON string.

ReleaseUnmanagedResources()

```
protected override void ReleaseUnmanagedResources()
```

Class LiteRTLMNativeHandle

Namespace: Uralstech.UAI.LiteRTLM

```
public abstract class LiteRTLMNativeHandle
```

Inheritance

object ← LiteRTLMNativeHandle

Derived

BenchmarkInfo, Conversation, ConversationConfig, ConversationOptionalArgs, DetokenizeResult, Engine, EngineSettings, InputData, JsonResponse, RepetitionPenaltyConfig, Responses, SamplerParams, Session, SessionConfig, ThinkingConfig, TokenUnion, TokenUnions, TokenizeResult

Properties

Native

```
protected nint Native { get; init; }
```

Property Value

nint

Methods

Dispose()

Destroys the native LiteRT LM object.

```
public void Dispose()
```

~LiteRTLMNativeHandle()

```
protected ~LiteRTLMNativeHandle()
```

ReleaseUnmanagedResources()

```
protected abstract void ReleaseUnmanagedResources()
```

ThrowIfDisposed()

```
protected void ThrowIfDisposed()
```

Operators

implicit operator nint(LiteRTLMNativeHandle?)

```
public static implicit operator nint(LiteRTLMNativeHandle? handle)
```

Parameters

`handle` LiteRTLMNativeHandle

Returns

nint

Class LiteRTLMNativeLogging

Namespace: Uralstech.UAI.LiteRTLM

```
public static class LiteRTLMNativeLogging
```

Inheritance

object ← LiteRTLMNativeLogging

Methods

SetMinLogLevel(LogSeverity)

Sets the minimum log level for the LiteRT LM library.

```
public static void SetMinLogLevel(LogSeverity level)
```

Parameters

`level` LogSeverity

Enum LogSeverity

Namespace: Uralstech.UAI.LiteRTLM

Represents the log severity / level.

```
public enum LogSeverity
```

Fields

`Debug = 1`

`Error = 4`

`Fatal = 5`

`Info = 2`

`Silent = 1000`

`Verbose = 0`

`Warning = 3`

Class RepetitionPenaltyConfig

Namespace: Uralstech.UAI.LiteRTLM

```
public sealed class RepetitionPenaltyConfig : LiteRTLMNativeHandle
```

Inheritance

object ← LiteRTLMNativeHandle ← RepetitionPenaltyConfig

Inherited Members

LiteRTLMNativeHandle.Dispose()

Constructors

RepetitionPenaltyConfig()

Creates a managed wrapper around a LiteRT LM repetition penalty configuration. The caller is responsible for disposing the wrapper using Dispose().

```
public RepetitionPenaltyConfig()
```

Exceptions

InvalidOperationException

Thrown if the native object could not be created.

Methods

ReleaseUnmanagedResources()

```
protected override void ReleaseUnmanagedResources()
```

SetFrequencyPenalty(float)

```
public void SetFrequencyPenalty(float frequencyPenalty)
```

Parameters

`frequencyPenalty` float

A scalar subtracted from a token's logit scaled by previous appearances.

SetPresencePenalty(float)

Sets the presence penalty for the repetition penalty config.

```
public void SetPresencePenalty(float presencePenalty)
```

Parameters

`presencePenalty` float

A scalar subtracted from a logit if a token has appeared at least once.

SetRepetitionPenalty(float)

Sets the repetition penalty for the repetition penalty config.

```
public void SetRepetitionPenalty(float repetitionPenalty)
```

Parameters

`repetitionPenalty` float

A multiplicative penalty for any token already generated.

SetWindowSize(int)

Sets the window size for the repetition penalty config.

```
public void SetWindowSize(int windowSize)
```

Parameters

`windowSize` int

The maximum number of recent tokens to consider.

Class Responses

Namespace: Uralstech.UAI.LiteRTLM

```
public sealed class Responses : LiteRTLMNativeHandle
```

Inheritance

object ← LiteRTLMNativeHandle ← Responses

Inherited Members

LiteRTLMNativeHandle.Dispose()

Properties

Count

Gets the number of elements in the collection.

```
public int Count { get; }
```

Property Value

int

The number of elements in the collection.

this[int]

Gets the element at the specified index in the read-only list.

```
public Responses.Candidate this[int index] { get; }
```

Parameters

`index` int

The zero-based index of the element to get.

Property Value

Responses.Candidate

The element at the specified index in the read-only list.

Methods

GetEnumerator()

Returns an enumerator that iterates through the collection.

```
public IEnumerator<Responses.Candidate> GetEnumerator()
```

Returns

IEnumerator<Responses.Candidate>

An enumerator that can be used to iterate through the collection.

Remarks

The returned enumerator is only valid for the lifetime of this Responses object.

ReleaseUnmanagedResources()

```
protected override void ReleaseUnmanagedResources()
```

Class Responses.Candidate

Namespace: Uralstech.UAI.LiteRTLM

```
public record Responses.Candidate
```

Inheritance

object ← Responses.Candidate

Constructors

Candidate(string, bool, float, bool, int, bool, float[]?)

```
public Candidate(string Text, bool HasScore, float Score, bool HasTokenLength, int
TokenLength, bool HasTokenScores, float[]? TokenScores)
```

Parameters

`Text` string

The response text.

`HasScore` bool

Does the response have a valid Score?

`Score` float

The response score.

`HasTokenLength` bool

Does the response have a valid TokenLength?

`TokenLength` int

The token length of this response.

`HasTokenScores` bool

Does the response have a valid TokenScores?

`TokenScores` float[]

The token scores for this response.

Properties

HasScore

Does the response have a valid Score?

```
public bool HasScore { get; init; }
```

Property Value

bool

HasTokenLength

Does the response have a valid TokenLength?

```
public bool HasTokenLength { get; init; }
```

Property Value

bool

HasTokenScores

Does the response have a valid TokenScores?

```
public bool HasTokenScores { get; init; }
```

Property Value

bool

Score

The response score.

```
public float Score { get; init; }
```

Property Value

float

Text

The response text.

```
public string Text { get; init; }
```

Property Value

string

TokenLength

The token length of this response.

```
public int TokenLength { get; init; }
```

Property Value

int

TokenScores

The token scores for this response.

```
public float[]? TokenScores { get; init; }
```

Property Value

float[]

Class SamplerParams

Namespace: Uralstech.UAI.LiteRTLM

```
public sealed class SamplerParams : LiteRTLMNativeHandle
```

Inheritance

object ← LiteRTLMNativeHandle ← SamplerParams

Inherited Members

LiteRTLMNativeHandle.Dispose()

Constructors

SamplerParams(SamplerType)

Creates a managed wrapper around LiteRT LM sampler parameters for a specific sampler type. The caller is responsible for disposing the wrapper using Dispose().

```
public SamplerParams(SamplerType samplerType)
```

Parameters

`samplerType` SamplerType

The sampler type to use.

Exceptions

InvalidOperationException

Thrown if the native object could not be created.

Methods

ReleaseUnmanagedResources()

```
protected override void ReleaseUnmanagedResources()
```

SetSeed(int)

Sets the seed.

```
public void SetSeed(int seed)
```

Parameters

`seed` int

SetTemperature(float)

Sets the temperature.

```
public void SetTemperature(float temperature)
```

Parameters

`temperature` float

SetTopK(int)

Sets the top-k value.

```
public void SetTopK(int topK)
```

Parameters

`topK` int

SetTopP(float)

Sets the top-p value.

```
public void SetTopP(float topP)
```

Parameters

`topP` float

Enum SamplerType

Namespace: Uralstech.UAI.LiteRTLM

Represents the type of sampler.

```
public enum SamplerType
```

Fields

`Greedy = 3`

Pick the token with maximum logit (i.e., argmax).

`TopK = 1`

Probabilistically pick among the top k tokens.

`TopP = 2`

Probabilistically pick among the tokens such that the sum is greater than or equal to p tokens after first performing top-k sampling.

Class Session

Namespace: Uralstech.UAI.LiteRTLM

```
public sealed class Session : LiteRTLMNativeHandle
```

Inheritance

object ← LiteRTLMNativeHandle ← Session

Inherited Members

LiteRTLMNativeHandle.Dispose()

Methods

CancelProcess()

Cancels the current processing in the session.

```
public void CancelProcess()
```

GenerateContent(ReadOnlySpan<InputData>)

Generates content from the input prompt.

```
public Responses? GenerateContent(ReadOnlySpan<InputData> inputs)
```

Parameters

`inputs` ReadOnlySpan<InputData>

An array of input wrappers representing multimodal input.

Returns

Responses

The responses wrapper, or null on failure. The caller is responsible for disposing the returned wrapper.

GenerateContentStream(ReadOnlySpan<InputData>, StreamCallback)

Generates content from the input prompt and streams the response through a callback. This is a non-blocking call and invokes the callback from a background thread for each chunk.

```
public int GenerateContentStream(ReadOnlySpan<InputData> inputs, StreamCallback callback)
```

Parameters

`inputs` ReadOnlySpan<InputData>

An array of input wrappers representing multimodal input.

`callback` StreamCallback

The callback function (StreamCallback) that receives response chunks.

Returns

int

0 on success, non-zero on failure to start the stream.

GetBenchmarkInfo()

Retrieves benchmark information for the session. The caller is responsible for disposing the returned wrapper.

```
public BenchmarkInfo? GetBenchmarkInfo()
```

Returns

BenchmarkInfo

The benchmark information wrapper, or null on failure.

ReleaseUnmanagedResources()

```
protected override void ReleaseUnmanagedResources()
```

RunDecode()

Starts decoding for the model to predict a response based on the input prompt or query added after calling RunPrefill(ReadOnlySpan<InputData>). This is a blocking call and returns when decoding is complete.

```
public Responses? RunDecode()
```

Returns

Responses

The responses wrapper, or null on failure. The caller is responsible for disposing the returned wrapper.

RunDecodeAsync(StreamCallback)

Starts decoding for the model to predict a response based on the input prompt or query added after calling RunPrefill(ReadOnlySpan<InputData>). This is a non-blocking call that streams response chunks through a callback.

```
public int RunDecodeAsync(StreamCallback callback)
```

Parameters

`callback` StreamCallback

The callback function (StreamCallback) that receives response chunks.

Returns

int

0 on success, non-zero on failure.

RunPrefill(ReadOnlySpan<InputData>)

Adds the input prompt or query to the model and starts the prefill process. This is a blocking call and returns when prefill is complete.

```
public int RunPrefill(ReadOnlySpan<InputData> inputs)
```

Parameters

`inputs` ReadOnlySpan<InputData>

An array of input wrappers representing multimodal input.

Returns

int

0 on success, non-zero on failure.

RunTextScoring(string[], bool)

Scores the target text after the prefill process is complete.

```
public Responses? RunTextScoring(string[] targetText, bool storeTokenLengths)
```

Parameters

`targetText` string[]

An array of target text strings to score.

`storeTokenLengths` bool

Whether to store the token lengths of the target texts in the responses.

Returns

Responses

The responses wrapper, or null on failure. The caller is responsible for disposing the returned wrapper.

Class SessionConfig

Namespace: Uralstech.UAI.LiteRTLM

```
public sealed class SessionConfig : LiteRTLMNativeHandle
```

Inheritance

object ← LiteRTLMNativeHandle ← SessionConfig

Inherited Members

LiteRTLMNativeHandle.Dispose()

Constructors

SessionConfig()

Creates a managed wrapper around a LiteRT LM session configuration. The caller is responsible for disposing the wrapper using Dispose().

```
public SessionConfig()
```

Exceptions

InvalidOperationException

Thrown if the native object could not be created.

Methods

ReleaseUnmanagedResources()

```
protected override void ReleaseUnmanagedResources()
```

SetApplyPromptTemplate(bool)

Sets whether to apply the prompt template for this session.

```
public void SetApplyPromptTemplate(bool applyPromptTemplate)
```

Parameters

`applyPromptTemplate` bool

Whether to apply the prompt template.

SetAudioLoraPath(string)

Sets the path to the audio LoRA weights file.

```
public int SetAudioLoraPath(string audioLoraPath)
```

Parameters

`audioLoraPath` string

The path to the audio LoRA weights file.

Returns

int

0 on success, non-zero on failure.

SetLoraPath(string)

Sets the path to the LoRA weights file.

```
public int SetLoraPath(string loraPath)
```

Parameters

`loraPath` string

The path to the text LoRA weights file.

Returns

int

0 on success, non-zero on failure.

SetMaxOutputTokens(int)

Sets the maximum number of output tokens per decode step for this session.

```
public void SetMaxOutputTokens(int maxOutputTokens)
```

Parameters

`maxOutputTokens` int

The maximum number of output tokens.

SetSamplerParams(SamplerParams)

Sets the sampler parameters for this session configuration.

```
public void SetSamplerParams(SamplerParams samplerParams)
```

Parameters

`samplerParams` [SamplerParams]

The sampler parameters to use.

Delegate StreamCallback

Namespace: Uralstech.UAI.LiteRTLM

Callback for streaming responses.

```
public delegate void StreamCallback(StreamChunk chunk)
```

Parameters

`chunk` StreamChunk

The stream chunk object. It's only valid for the duration of the call.

Class StreamChunk

Namespace: Uralstech.UAI.LiteRTLM

```
public sealed class StreamChunk
```

Inheritance

object ← StreamChunk

Constructors

StreamChunk(nint)

```
public StreamChunk(nint native)
```

Parameters

`native` nint

Methods

GetError()

Gets the error message associated with this chunk, if any.

```
public string? GetError()
```

Returns

string

Remarks

Returns null if there is no error.

GetText()

Gets the text content of the chunk.

```
public string? GetText()
```

Returns

string

Returns null if there is no text content in this chunk (e.g. if it is an error or metadata-only chunk).

IsFinal()

Returns true if this is the final chunk of the stream.

```
public bool IsFinal()
```

Returns

bool

Class ThinkingConfig

Namespace: Uralstech.UAI.LiteRTLM

```
public sealed class ThinkingConfig : LiteRTLMNativeHandle
```

Inheritance

object ← LiteRTLMNativeHandle ← ThinkingConfig

Inherited Members

LiteRTLMNativeHandle.Dispose()

Constructors

ThinkingConfig()

Creates a managed wrapper around a LiteRT LM thinking configuration. The caller is responsible for disposing the wrapper using Dispose().

```
public ThinkingConfig()
```

Exceptions

InvalidOperationException

Thrown if the native object could not be created.

Methods

ReleaseUnmanagedResources()

```
protected override void ReleaseUnmanagedResources()
```

SetEnableThinking(bool)

Sets whether thinking/reasoning generation is enabled.

```
public void SetEnableThinking(bool enableThinking)
```

Parameters

`enableThinking` bool

Whether thinking is enabled.

SetThinkingTokenBudget(int)

Sets the thinking token budget.

```
public void SetThinkingTokenBudget(int thinkingTokenBudget)
```

Parameters

`thinkingTokenBudget` int

Budget for token-by-token reasoning generation (-1 for infinite).

Class TokenUnion

Namespace: Uralstech.UAI.LiteRTLM

```
public sealed class TokenUnion : LiteRTLMNativeHandle
```

Inheritance

object ← LiteRTLMNativeHandle ← TokenUnion

Inherited Members

LiteRTLMNativeHandle.Dispose()

Methods

GetIds(out int[]?)

Returns the token ids from a token union.

```
public int GetIds(out int[]? tokenIds)
```

Parameters

`tokenIds` int[]

The token IDs.

Returns

int

0 on success, non-zero if the token union does not contain token ids.

GetString()

Returns the string value from a token union.

```
public string? GetString()
```

Returns

string

The string value, or null if the token union does not contain a string value.

GetUnionType()

Returns the type of the token union.

```
public TokenUnionType GetUnionType()
```

Returns

TokenUnionType

The type of the token union.

ReleaseUnmanagedResources()

```
protected override void ReleaseUnmanagedResources()
```

Enum TokenUnionType

Namespace: Uralstech.UAI.LiteRTLM

Represents the type of a TokenUnion.

```
public enum TokenUnionType
```

Fields

`Ids = 1`

`String = 0`

Class TokenUnions

Namespace: Uralstech.UAI.LiteRTLM

```
public sealed class TokenUnions : LiteRTLMNativeHandle
```

Inheritance

object ← LiteRTLMNativeHandle ← TokenUnions

Inherited Members

LiteRTLMNativeHandle.Dispose()

Methods

GetTokenUnions()

Gets the token unions in this collection.

```
public TokenUnion[] GetTokenUnions()
```

Returns

TokenUnion[]

The token union wrappers in the collection. The caller is responsible for disposing each returned wrapper.

ReleaseUnmanagedResources()

```
protected override void ReleaseUnmanagedResources()
```

Class TokenizeResult

Namespace: Uralstech.UAI.LiteRTLM

```
public sealed class TokenizeResult : LiteRTLMNativeHandle
```

Inheritance

object ← LiteRTLMNativeHandle ← TokenizeResult

Inherited Members

LiteRTLMNativeHandle.Dispose()

Methods

AsReadOnlySpan()

Returns a span of the token ids.

```
public ReadOnlySpan<int> AsReadOnlySpan()
```

Returns

ReadOnlySpan<int>

Remarks

The returned span is only valid for the lifetime of this TokenizeResult object.

CopyTo(Span<int>)

Copies the token ids into `span`.

```
public long CopyTo(Span<int> span)
```

Parameters

`span` Span<int>

Returns

long

The number of copied elements.

ReleaseUnmanagedResources()

```
protected override void ReleaseUnmanagedResources()
```

Namespace Uralstech.UAI.LiteRTLM.Native

Classes

NativeAPI

NativeAPI.BenchmarkInfo

NativeAPI.Conversation

NativeAPI.ConversationConfig

NativeAPI.ConversationOptionalArgs

NativeAPI.DetokenizeResult

NativeAPI.Engine

NativeAPI.EngineSettings

NativeAPI.InputData

NativeAPI.JsonResponse

NativeAPI.RepetitionPenaltyConfig

NativeAPI.Responses

NativeAPI.SamplerParams

NativeAPI.Session

NativeAPI.SessionConfig

NativeAPI.StreamChunk

NativeAPI.ThinkingConfig

NativeAPI.TokenUnion

NativeAPI.TokenUnions

NativeAPI.TokenizeResult

Delegates

NativeAPI.StreamCallback

Callback for streaming responses.

Class NativeAPI

Namespace: Uralstech.[UAI](UAI).LiteRTLM.Native

```
public static class NativeAPI
```

Inheritance

object ← NativeAPI

Fields

LibLiteRTLM

```
public const string LibLiteRTLM = "litert-lm"
```

Field Value

string

Methods

litert_lm_set_min_log_level(LogSeverity)

Sets the minimum log level for the LiteRT LM library.

```
public static extern void litert_lm_set_min_log_level(LogSeverity level)
```

Parameters

`level` LogSeverity

Class NativeAPI.BenchmarkInfo

Namespace: Uralstech.UAI.LiteRTLM.Native

```
public static class NativeAPI.BenchmarkInfo
```

Inheritance

object ← NativeAPI.BenchmarkInfo

Methods

litert_lm_benchmark_info_delete(nint)

Destroys a LiteRT LM Benchmark Info object.

```
public static extern void litert_lm_benchmark_info_delete(nint benchmarkInfo)
```

Parameters

`benchmarkInfo` nint

The benchmark info to destroy.

litert_lm_benchmark_info_get_decode_token_count_at(nint, int)

Returns the decode token count at a given turn index.

```
public static extern int litert_lm_benchmark_info_get_decode_token_count_at(nint
benchmarkInfo, int index)
```

Parameters

`benchmarkInfo` nint

The benchmark info object.

`index` int

The index of the decode turn.

Returns

int

The decode token count.

litert_lm_benchmark_info_get_decode_tokens_per_sec_at(nint, int)

Returns the decode tokens per second at a given turn index.

```
public static extern double litert_lm_benchmark_info_get_decode_tokens_per_sec_at(nint
benchmarkInfo, int index)
```

Parameters

`benchmarkInfo` nint

The benchmark info object.

`index` int

The index of the decode turn.

Returns

double

The decode tokens per second.

litert_lm_benchmark_info_get_num_decode_turns(nint)

Returns the number of decode turns.

```
public static extern int litert_lm_benchmark_info_get_num_decode_turns(nint benchmarkInfo)
```

Parameters

`benchmarkInfo` nint

The benchmark info object.

Returns

int

The number of decode turns.

litert_lm_benchmark_info_get_num_prefill_turns(nint)

Returns the number of prefill turns.

```
public static extern int litert_lm_benchmark_info_get_num_prefill_turns(nint benchmarkInfo)
```

Parameters

`benchmarkInfo` nint

The benchmark info object.

Returns

int

The number of prefill turns.

litert_lm_benchmark_info_get_prefill_token_count_at(nint, int)

Returns the prefill token count at a given turn index.

```
public static extern int litert_lm_benchmark_info_get_prefill_token_count_at(nint
benchmarkInfo, int index)
```

Parameters

`benchmarkInfo` nint

The benchmark info object.

`index` int

The index of the prefill turn.

Returns

int

The prefill token count.

litert_lm_benchmark_info_get_prefill_tokens_per_sec_at(nint, int)

Returns the prefill tokens per second at a given turn index.

```
public static extern double litert_lm_benchmark_info_get_prefill_tokens_per_sec_at(nint
benchmarkInfo, int index)
```

Parameters

`benchmarkInfo` nint

The benchmark info object.

`index` int

The index of the prefill turn.

Returns

double

The prefill tokens per second.

litert_lm_benchmark_info_get_time_to_first_token(nint)

Returns the time to the first token in seconds.

```
public static extern double litert_lm_benchmark_info_get_time_to_first_token(nint
benchmarkInfo)
```

Parameters

`benchmarkInfo` nint

The benchmark info object.

Returns

double

The time to the first token in seconds.

Remarks

Note that the first time to token doesn't include the time for initialization. It is the sum of the prefill time for the first turn and the time spent for decoding the first token.

litert_lm_benchmark_info_get_total_init_time_in_second(nint)

Returns the total initialization time in seconds.

```
public static extern double litert_lm_benchmark_info_get_total_init_time_in_second(nint
benchmarkInfo)
```

Parameters

`benchmarkInfo` nint

The benchmark info object.

Returns

double

The total initialization time in seconds.

Class NativeAPI.Conversation

Namespace: Uralstech.UAI.LiteRTLM.Native

```
public static class NativeAPI.Conversation
```

Inheritance

object ← NativeAPI.Conversation

Methods

litert_lm_conversation_cancel_process(nint)

Cancels the ongoing inference process, for asynchronous inference.

```
public static extern void litert_lm_conversation_cancel_process(nint conversation)
```

Parameters

`conversation` nint

The conversation to cancel the inference for.

litert_lm_conversation_clone(nint)

Clones a LiteRT LM Conversation, duplicating its prefilled state. The caller is responsible for destroying the cloned conversation using litert_lm_conversation_delete(nint).

```
public static extern nint litert_lm_conversation_clone(nint conversation)
```

Parameters

`conversation` nint

The conversation to clone.

Returns

nint

A pointer to the cloned conversation, or nint.Zero on failure.

litert_lm_conversation_create(nint, nint)

Creates a LiteRT LM Conversation. The caller is responsible for destroying the conversation using litert_lm_conversation_delete(nint).

```
public static extern nint litert_lm_conversation_create(nint engine, nint config)
```

Parameters

`engine` nint

The engine to create the conversation from.

`config` nint

The conversation config to use. If nint.Zero, the default config will be used.

Returns

nint

A pointer to the created conversation, or nint.Zero on failure.

litert_lm_conversation_delete(nint)

Destroys a LiteRT LM Conversation.

```
public static extern void litert_lm_conversation_delete(nint conversation)
```

Parameters

`conversation` nint

The conversation to destroy.

litert_lm_conversation_get_benchmark_info(nint)

Retrieves the benchmark information from the conversation. The caller is responsible for destroying the benchmark info using [litert_lm_benchmark_info_delete(nint)].

```
public static extern nint litert_lm_conversation_get_benchmark_info(nint conversation)
```

Parameters

`conversation` nint

The conversation to get the benchmark info from.

Returns

nint

A pointer to the benchmark info, or nint.Zero on failure.

litert_lm_conversation_get_token_count(nint)

Gets the number of tokens in the conversation KV Cache (prefill + decode).

```
public static extern int litert_lm_conversation_get_token_count(nint conversation)
```

Parameters

`conversation` nint

Returns

int

The number of tokens, or a negative value on failure.

litert_lm_conversation_render_message_to_string(nint, string)

Renders the message into a string according to the template.

```
public static extern nint litert_lm_conversation_render_message_to_string(nint conversation,
string messageJson)
```

Parameters

`conversation` nint

The conversation instance.

`messageJson` string

A JSON string representing the message to render.

Returns

nint

A pointer to the rendered string, or nint.Zero on failure. The returned string is owned by the Conversation object and is valid until the next call to this function or until the conversation is deleted.

Remarks

This function does not need to be called for actual message sending, as the [litert_lm_conversation_send_message(nint, string, string?, nint)] and [litert_lm_conversation_send_message_stream(nint, string, string?, nint, nint, nint)] functions will handle rendering internally.

litert_lm_conversation_render_preface_to_string(nint)

Renders the preface into a string according to the template.

```
public static extern nint litert_lm_conversation_render_preface_to_string(nint conversation)
```

Parameters

`conversation` nint

The conversation instance.

Returns

nint

A pointer to the rendered string, or nint.Zero on failure. The returned string is owned by the Conversation object and is valid until the next call to this function or until the conversation is deleted.

litert_lm_conversation_send_message(nint, string, string?, nint)

Sends a message to the conversation and returns the response. This is a blocking call.

```
public static extern nint litert_lm_conversation_send_message(nint conversation, string
messageJson, string? extraContext, nint optionalArgs)
```

Parameters

`conversation` nint

The conversation to use.

`messageJson` string

A JSON string representing the message to send.

`extraContext` string

A JSON string representing the extra context to use.

`optionalArgs` nint

A pointer to the optional arguments to use.

Returns

nint

A pointer to the JSON Response, or nint.Zero on failure. The caller is responsible for deleting the response using [litert_lm_json_response_delete(nint)](#).

litert_lm_conversation_send_message_stream(nint, string, string?, nint, nint, nint)

Sends a message to the conversation and streams the response via a callback. This is a non-blocking call that will invoke the callback from a background thread for each chunk.

```
public static extern int litert_lm_conversation_send_message_stream(nint conversation,
string messageJson, string? extraContext, nint optionalArgs, nint callback,
nint callbackData)
```

Parameters

`conversation` nint

The conversation to use.

`messageJson` string

A JSON string representing the message to send.

`extraContext` string

A JSON string representing the extra context to use.

`optionalArgs` nint

A pointer to the optional arguments to use.

`callback` nint

The callback function ([NativeAPI.StreamCallback]()) to receive response chunks.

`callbackData` nint

A pointer to user data that will be passed to the callback.

Returns

int

0 on success, non-zero on failure to start the stream.

Class NativeAPI.ConversationConfig

Namespace: Uralstech.UAI.LiteRTLM.Native

```
public static class NativeAPI.ConversationConfig
```

Inheritance

object ← NativeAPI.ConversationConfig

Methods

litert_lm_conversation_config_create()

Creates a LiteRT LM Conversation Config. The caller is responsible for destroying the config using litert_lm_conversation_config_delete(nint).

```
public static extern nint litert_lm_conversation_config_create()
```

Returns

nint

A pointer to the created config, or nint.Zero on failure.

litert_lm_conversation_config_delete(nint)

Destroys a LiteRT LM Conversation Config.

```
public static extern void litert_lm_conversation_config_delete(nint config)
```

Parameters

`config` nint

The config to destroy.

litert_lm_conversation_config_set_enable_constrained_decoding(nint, bool)

Sets whether to enable constrained decoding for this conversation config.

```
public static extern void litert_lm_conversation_config_set_enable_constrained_decoding(nint
config, bool enableConstrainedDecoding)
```

Parameters

`config` nint

The config to modify.

`enableConstrainedDecoding` bool

Whether to enable constrained decoding.

litert_lm_conversation_config_set_extra_context(nint, string)

Sets the extra context for the conversation preface.

```
public static extern void litert_lm_conversation_config_set_extra_context(nint config,
string extraContextJson)
```

Parameters

`config` nint

The config to modify.

`extraContextJson` string

A JSON string representing the extra context object.

litert_lm_conversation_config_set_filter_channel_content_from_kv_cache(nint, bool)

Sets whether to filter channel content from the KV cache.

```
public static extern void
litert_lm_conversation_config_set_filter_channel_content_from_kv_cache(nint config, bool
filterChannelContentFromKvCache)
```

Parameters

`config` nint

The config to modify.

`filterChannelContentFromKvCache` bool

Whether to filter channel content.

litert_lm_conversation_config_set_messages(nint, string)

Sets the initial messages for this conversation config.

```
public static extern void litert_lm_conversation_config_set_messages(nint config,
string messagesJson)
```

Parameters

`config` nint

The config to modify.

`messagesJson` string

The initial messages in JSON array format.

litert_lm_conversation_config_set_session_config(nint, nint)

Sets the session config for this conversation config.

```
public static extern void litert_lm_conversation_config_set_session_config(nint config,
nint sessionConfig)
```

Parameters

`config` nint

The config to modify.

`sessionConfig` nint

The session config to use.

litert_lm_conversation_config_set_stream_tool_calls(nint, bool, string)

Sets whether to stream tool call tokens.

```
public static extern void litert_lm_conversation_config_set_stream_tool_calls(nint config,
bool streamToolCalls, string channelName)
```

Parameters

`config` nint

The config to modify.

`streamToolCalls` bool

Whether to stream tool call tokens.

`channelName` string

The channel name to use for tool call tokens.

litert_lm_conversation_config_set_system_message(nint, string)

Sets the system message for this conversation config.

```
public static extern void litert_lm_conversation_config_set_system_message(nint config,
string systemMessageJson)
```

Parameters

`config` nint

The config to modify.

`systemMessageJson` string

The system message in JSON format.

litert_lm_conversation_config_set_thinking_config(nint, nint)

Sets the thinking config for this conversation config.

```
public static extern void litert_lm_conversation_config_set_thinking_config(nint config,
nint thinkingConfig)
```

Parameters

`config` nint

The config to modify.

`thinkingConfig` nint

The thinking config to set. If nint.Zero, clears any previously set thinking config.

litert_lm_conversation_config_set_tools(nint, string)

Sets the tools for this conversation config.

```
public static extern void litert_lm_conversation_config_set_tools(nint config,
string toolsJson)
```

Parameters

`config` nint

The config to modify.

`toolsJson` string

The tools description in JSON array format.

Class NativeAPI.ConversationOptionalArgs

Namespace: Uralstech.UAI.LiteRTLM.Native

```
public static class NativeAPI.ConversationOptionalArgs
```

Inheritance

object ← NativeAPI.ConversationOptionalArgs

Methods

litert_lm_conversation_optional_args_create()

Creates a LiteRT LM Conversation Optional Args. The caller is responsible for destroying the optional args using litert_lm_conversation_optional_args_delete(nint).

```
public static extern nint litert_lm_conversation_optional_args_create()
```

Returns

nint

A pointer to the created optional args, or nint.Zero on failure.

litert_lm_conversation_optional_args_delete(nint)

Destroys a LiteRT LM Conversation Optional Args.

```
public static extern void litert_lm_conversation_optional_args_delete(nint optionalArgs)
```

Parameters

`optionalArgs` nint

The optional args to destroy.

litert_lm_conversation_optional_args_set_max_output_tokens(nint, int)

Sets the maximum number of output tokens for the conversation optional args.

```
public static extern void litert_lm_conversation_optional_args_set_max_output_tokens(nint
optionalArgs, int maxOutputTokens)
```

Parameters

`optionalArgs` nint

The optional args to modify.

`maxOutputTokens` int

The maximum number of output tokens.

litert_lm_conversation_optional_args_set_repetition_penalty_config(nint, nint)

Sets the repetition penalty configuration for the conversation optional args.

```
public static extern void
litert_lm_conversation_optional_args_set_repetition_penalty_config(nint optionalArgs,
nint repetitionPenaltyConfig)
```

Parameters

`optionalArgs` nint

The optional args to modify.

`repetitionPenaltyConfig` nint

The repetition penalty config to set. If nint.Zero, clears any previously set repetition penalty config.

litert_lm_conversation_optional_args_set_thinking_config(nint, nint)

Sets the thinking config for the conversation optional args.

```
public static extern void litert_lm_conversation_optional_args_set_thinking_config(nint
optionalArgs, nint thinkingConfig)
```

Parameters

`optionalArgs` nint

The optional args to modify.

`thinkingConfig` nint

The thinking config to set. If nint.Zero, clears any previously set thinking config.

litert_lm_conversation_optional_args_set_visual_token_budget(nint, int)

Sets the visual token budget for the conversation optional args.

```
public static extern void litert_lm_conversation_optional_args_set_visual_token_budget(nint
optionalArgs, int visualTokenBudget)
```

Parameters

`optionalArgs` nint

The optional args to modify.

`visualTokenBudget` int

The visual token budget.

Class NativeAPI.DetokenizeResult

Namespace: Uralstech.UAI.LiteRTLM.Native

```
public static class NativeAPI.DetokenizeResult
```

Inheritance

object ← NativeAPI.DetokenizeResult

Methods

litert_lm_detokenize_result_delete(nint)

Destroys a LiteRT LM Detokenize Result.

```
public static extern void litert_lm_detokenize_result_delete(nint result)
```

Parameters

`result` nint

The detokenize result to destroy.

litert_lm_detokenize_result_get_string(nint)

Returns the string from a detokenize result.

```
public static extern nint litert_lm_detokenize_result_get_string(nint result)
```

Parameters

`result` nint

The detokenize result.

Returns

nint

The detokenized string. The returned string is owned by the DetokenizeResult object and is valid only for its lifetime.

Class NativeAPI.Engine

Namespace: Uralstech.UAI.LiteRTLM.Native

```
public static class NativeAPI.Engine
```

Inheritance

object ← NativeAPI.Engine

Methods

litert_lm_engine_create(nint)

Creates a LiteRT LM Engine from the given settings. The caller is responsible for destroying the engine using litert_lm_engine_delete(nint).

```
public static extern nint litert_lm_engine_create(nint settings)
```

Parameters

`settings` nint

The engine settings.

Returns

nint

A pointer to the created engine, or nint.Zero on failure.

litert_lm_engine_create_session(nint, nint)

Creates a LiteRT LM Session. The caller is responsible for destroying the session using litert_lm_session_delete(nint).

```
public static extern nint litert_lm_engine_create_session(nint engine, nint config)
```

Parameters

`engine` nint

The engine to create the session from.

`config` nint

The session config of the session. If nint.Zero, use the default session config.

Returns

nint

A pointer to the created session, or nint.Zero on failure.

litert_lm_engine_delete(nint)

Destroys a LiteRT LM Engine.

```
public static extern void litert_lm_engine_delete(nint engine)
```

Parameters

`engine` nint

The engine to destroy.

litert_lm_engine_detokenize(nint, int[], nuint)

Detokenizes token ids using the engine's tokenizer.

```
public static extern nint litert_lm_engine_detokenize(nint engine, int[] tokens,
nuint numTokens)
```

Parameters

`engine` nint

The engine instance.

`tokens` int[]

An array of token ids to detokenize.

`numTokens` nuint

The number of token ids in the array.

Returns

nint

A pointer to the detokenize result, or nint.Zero on failure. The caller is responsible for deleting the result using litert_lm_detokenize_result_delete(nint).

litert_lm_engine_get_start_token(nint)

Returns the configured start token (BOS), if any.

```
public static extern nint litert_lm_engine_get_start_token(nint engine)
```

Parameters

`engine` nint

The engine instance.

Returns

nint

A pointer to the start token, or nint.Zero if none configured. The caller is responsible for deleting the result using litert_lm_token_union_delete(nint).

litert_lm_engine_get_stop_tokens(nint)

Returns the configured stop tokens (EOS).

```
public static extern nint litert_lm_engine_get_stop_tokens(nint engine)
```

Parameters

`engine` nint

The engine instance.

Returns

nint

A pointer to the stop tokens collection, or nint.Zero if none configured. The caller is responsible for deleting the result using [litert_lm_token_unions_delete(nint)]().

litert_lm_engine_tokenize(nint, string)

Tokenizes text using the engine's tokenizer.

```
public static extern nint litert_lm_engine_tokenize(nint engine, string text)
```

Parameters

`engine` nint

The engine instance.

`text` string

The UTF-8 string to tokenize.

Returns

nint

A pointer to the tokenize result, or nint.Zero on failure. The caller is responsible for deleting the result using [litert_lm_tokenize_result_delete(nint)]().

Class NativeAPI.EngineSettings

Namespace: Uralstech.UAI.LiteRTLM.Native

```
public static class NativeAPI.EngineSettings
```

Inheritance

object ← NativeAPI.EngineSettings

Methods

litert_lm_engine_settings_create(string, string, string?, string?)

Creates LiteRT LM Engine Settings. The caller is responsible for destroying the settings using litert_lm_engine_settings_delete(nint).

```
public static extern nint litert_lm_engine_settings_create(string modelPath, string
backendStr, string? visionBackendStr, string? audioBackendStr)
```

Parameters

`modelPath` string

The path to the model file.

`backendStr` string

The backend to use (e.g., "cpu", "gpu").

`visionBackendStr` string

The vision backend to use, or null if not set.

`audioBackendStr` string

The audio backend to use, or null if not set.

Returns

nint

A pointer to the created settings, or nint.Zero on failure.

litert_lm_engine_settings_create_from_raw_file_descriptor(int, string, string?, string?)

Creates LiteRT LM Engine Settings from a raw file descriptor. The engine takes ownership of the file descriptor and will close it when done. The caller is responsible for destroying the settings using [litert_lm_engine_settings_delete(nint)]().

```
public static extern nint litert_lm_engine_settings_create_from_raw_file_descriptor(int fd,
string backendStr, string? visionBackendStr, string? audioBackendStr)
```

Parameters

`fd` int

The file descriptor of the model.

`backendStr` string

The backend to use (e.g., "cpu", "gpu").

`visionBackendStr` string

The vision backend to use, or [null]() if not set.

`audioBackendStr` string

The audio backend to use, or [null]() if not set.

Returns

nint

A pointer to the created settings, or nint.Zero on failure.

litert_lm_engine_settings_delete(nint)

Destroys LiteRT LM Engine Settings.

```
public static extern void litert_lm_engine_settings_delete(nint settings)
```

Parameters

`settings` nint

The settings to destroy.

litert_lm_engine_settings_enable_benchmark(nint)

Enables benchmarking for the engine.

```
public static extern void litert_lm_engine_settings_enable_benchmark(nint settings)
```

Parameters

`settings` nint

The engine settings.

litert_lm_engine_settings_set_activation_data_type(nint, ActivationDataType)

Sets the activation data type.

```
public static extern void litert_lm_engine_settings_set_activation_data_type(nint settings,
ActivationDataType activationDataType)
```

Parameters

`settings` nint

The engine settings.

`activationDataType` [ActivationDataType]

The activation data type.

litert_lm_engine_settings_set_audio_lora_rank(nint, int)

Sets the Audio LoRA rank for the engine.

```
public static extern void litert_lm_engine_settings_set_audio_lora_rank(nint settings,
int loraRank)
```

Parameters

`settings` nint

The engine settings.

`loraRank` int

The Audio LoRA rank.

litert_lm_engine_settings_set_audio_num_threads(nint, int)

Sets the number of threads for the audio CPU backend.

```
public static extern void litert_lm_engine_settings_set_audio_num_threads(nint settings,
int numThreads)
```

Parameters

`settings` nint

The engine settings.

`numThreads` int

The number of threads.

litert_lm_engine_settings_set_cache_dir(nint, string)

Sets the cache directory for the engine.

```
public static extern void litert_lm_engine_settings_set_cache_dir(nint settings,
```

```
string cacheDir)
```

Parameters

`settings` nint

The engine settings.

`cacheDir` string

The cache directory.

litert_lm_engine_settings_set_enable_speculative_decoding(nint, bool)

Sets whether to enable speculative decoding.

```
public static extern void litert_lm_engine_settings_set_enable_speculative_decoding(nint
settings, bool enableSpeculativeDecoding)
```

Parameters

`settings` nint

The engine settings.

`enableSpeculativeDecoding` bool

Whether to enable speculative decoding.

litert_lm_engine_settings_set_litert_dispatch_lib_dir(nint, string)

Sets the LiteRT dispatch library directory for NPU backend.

```
public static extern void litert_lm_engine_settings_set_litert_dispatch_lib_dir(nint
settings, string libDir)
```

Parameters

`settings` nint

The engine settings.

`libDir` string

The dispatch library directory.

litert_lm_engine_settings_set_lora_rank(nint, int)

Sets the LoRA rank for the engine.

```
public static extern void litert_lm_engine_settings_set_lora_rank(nint settings,
int loraRank)
```

Parameters

`settings` nint

The engine settings.

`loraRank` int

The LoRA rank.

litert_lm_engine_settings_set_max_num_images(nint, int)

Sets the maximum number of images for the engine. This is only used for the legacy implementation of the engine.

```
public static extern void litert_lm_engine_settings_set_max_num_images(nint settings,
int maxNumImages)
```

Parameters

`settings` nint

The engine settings.

`maxNumImages` int

The maximum number of images.

litert_lm_engine_settings_set_max_num_tokens(nint, int)

Sets the maximum number of tokens for the engine.

```
public static extern void litert_lm_engine_settings_set_max_num_tokens(nint settings,
int maxNumTokens)
```

Parameters

`settings` nint

The engine settings.

`maxNumTokens` int

The maximum number of tokens.

litert_lm_engine_settings_set_num_decode_tokens(nint, int)

Sets the number of decode tokens for benchmarking.

```
public static extern void litert_lm_engine_settings_set_num_decode_tokens(nint settings,
int numDecodeTokens)
```

Parameters

`settings` nint

The engine settings.

`numDecodeTokens` int

The number of decode tokens.

litert_lm_engine_settings_set_num_prefill_tokens(nint, int)

Sets the number of prefill tokens for benchmarking.

```
public static extern void litert_lm_engine_settings_set_num_prefill_tokens(nint settings,
int numPrefillTokens)
```

Parameters

`settings` nint

The engine settings.

`numPrefillTokens` int

The number of prefill tokens.

litert_lm_engine_settings_set_num_threads(nint, int)

Sets the number of threads for the CPU backend.

```
public static extern void litert_lm_engine_settings_set_num_threads(nint settings,
int numThreads)
```

Parameters

`settings` nint

The engine settings.

`numThreads` int

The number of threads.

litert_lm_engine_settings_set_parallel_file_section_loading(nint, bool)

Sets whether the engine should load different sections of the litertlm file in parallel. Defaults to true.

```
public static extern void litert_lm_engine_settings_set_parallel_file_section_loading(nint
settings, bool parallelFileSectionLoading)
```

Parameters

`settings` nint

The engine settings.

`parallelFileSectionLoading` bool

Whether to load in parallel.

litert_lm_engine_settings_set_prefill_chunk_size(nint, int)

Sets the prefill chunk size for the engine. Only applicable for CPU backend with dynamic models.

```
public static extern void litert_lm_engine_settings_set_prefill_chunk_size(nint settings,
int prefillChunkSize)
```

Parameters

`settings` nint

The engine settings.

`prefillChunkSize` int

The prefill chunk size.

litert_lm_engine_settings_set_supported_audio_lora_ranks(nint, int[], nuint)

Sets the supported Audio LoRA ranks for the engine.

```
public static extern int litert_lm_engine_settings_set_supported_audio_lora_ranks(nint
settings, int[] loraRanks, nuint numRanks)
```

Parameters

`settings` nint

The engine settings.

`loraRanks` int[]

An array of supported Audio LoRA ranks.

`numRanks` nuint

The number of ranks in the array.

Returns

int

0 on success, non-zero on failure.

litert_lm_engine_settings_set_supported_lora_ranks(nint, int[], nuint)

Sets the supported LoRA ranks for the engine.

```
public static extern int litert_lm_engine_settings_set_supported_lora_ranks(nint settings,
int[] loraRanks, nuint numRanks)
```

Parameters

`settings` nint

The engine settings.

`loraRanks` int[]

An array of supported LoRA ranks.

`numRanks` nuint

The number of ranks in the array.

Returns

int

0 on success, non-zero on failure.

Class NativeAPI.InputData

Namespace: Uralstech.UAI.LiteRTLM.Native

```
public static class NativeAPI.InputData
```

Inheritance

object ← NativeAPI.InputData

Methods

litert_lm_input_data_create(InputDataType, nint, nuint)

Creates a LiteRT LM Input Data. The caller is responsible for destroying the input data using litert_lm_input_data_delete(nint).

```
public static extern nint litert_lm_input_data_create(InputDataType type, nint data,
nuint size)
```

Parameters

`type` InputDataType

The type of the input data.

`data` nint

The data pointer. The data is copied internally.

`size` nuint

The size of the data in bytes.

Returns

nint

A pointer to the created input data, or nint.Zero on failure.

litert_lm_input_data_delete(nint)

Destroys a LiteRT LM Input Data.

```
public static extern void litert_lm_input_data_delete(nint inputData)
```

Parameters

`inputData` nint

The input data to destroy.

Class NativeAPI.JsonResponse

Namespace: Uralstech.UAI.LiteRTLM.Native

```
public static class NativeAPI.JsonResponse
```

Inheritance

object ← NativeAPI.JsonResponse

Methods

litert_lm_json_response_delete(nint)

Destroys a LiteRT LM Json Response object.

```
public static extern void litert_lm_json_response_delete(nint response)
```

Parameters

`response` nint

The response to destroy.

litert_lm_json_response_get_string(nint)

Returns the JSON response string from a response object.

```
public static extern nint litert_lm_json_response_get_string(nint response)
```

Parameters

`response` nint

The response object.

Returns

nint

The response JSON string. The returned string is owned by the JsonResponse object and is valid only for its lifetime. Returns nint.Zero if `response` is nint.Zero.

Class NativeAPI.RepetitionPenaltyConfig

Namespace: Uralstech.UAI.LiteRTLM.Native

```
public static class NativeAPI.RepetitionPenaltyConfig
```

Inheritance

object ← NativeAPI.RepetitionPenaltyConfig

Methods

litert_lm_repetition_penalty_config_create()

Creates a LiteRT LM Repetition Penalty Config. The caller is responsible for destroying the config using litert_lm_repetition_penalty_config_delete(nint).

```
public static extern nint litert_lm_repetition_penalty_config_create()
```

Returns

nint

A pointer to the created config, or nint.Zero on failure.

litert_lm_repetition_penalty_config_delete(nint)

Destroys a LiteRT LM Repetition Penalty Config.

```
public static extern void litert_lm_repetition_penalty_config_delete(nint config)
```

Parameters

`config` nint

The config to destroy.

litert_lm_repetition_penalty_config_set_frequency_penalty(nint, float)

```
public static extern void litert_lm_repetition_penalty_config_set_frequency_penalty(nint
config, float frequencyPenalty)
```

Parameters

`config` nint

The config to modify.

`frequencyPenalty` float

A scalar subtracted from a token's logit scaled by previous appearances.

litert_lm_repetition_penalty_config_set_presence_penalty(nint, float)

Sets the presence penalty for the repetition penalty config.

```
public static extern void litert_lm_repetition_penalty_config_set_presence_penalty(nint
config, float presencePenalty)
```

Parameters

`config` nint

The config to modify.

`presencePenalty` float

A scalar subtracted from a logit if a token has appeared at least once.

litert_lm_repetition_penalty_config_set_repetition_penalty(nint, float)

Sets the repetition penalty for the repetition penalty config.

```
public static extern void litert_lm_repetition_penalty_config_set_repetition_penalty(nint
config, float repetitionPenalty)
```

Parameters

`config` nint

The config to modify.

`repetitionPenalty` float

A multiplicative penalty for any token already generated.

litert_lm_repetition_penalty_config_set_window_size(nint, int)

Sets the window size for the repetition penalty config.

```
public static extern void litert_lm_repetition_penalty_config_set_window_size(nint config,
int windowSize)
```

Parameters

`config` nint

The config to modify.

`windowSize` int

The maximum number of recent tokens to consider.

Class NativeAPI.Responses

Namespace: Uralstech.UAI.LiteRTLM.Native

```
public static class NativeAPI.Responses
```

Inheritance

object ← NativeAPI.Responses

Methods

litert_lm_responses_delete(nint)

Destroys a LiteRT LM Responses object.

```
public static extern void litert_lm_responses_delete(nint responses)
```

Parameters

`responses` nint

The responses to destroy.

litert_lm_responses_get_num_candidates(nint)

Returns the number of response candidates.

```
public static extern int litert_lm_responses_get_num_candidates(nint responses)
```

Parameters

`responses` nint

The responses object.

Returns

int

The number of candidates.

litert_lm_responses_get_num_token_scores_at(nint, int)

Returns the number of tokens for which scores are present at a given index.

```
public static extern int litert_lm_responses_get_num_token_scores_at(nint responses,
int index)
```

Parameters

`responses` nint

The responses object.

`index` int

The index of the response.

Returns

int

The number of token scores. Returns 0 if index is out of bounds or no token scores are present.

litert_lm_responses_get_response_text_at(nint, int)

Returns the response text at a given index.

```
public static extern nint litert_lm_responses_get_response_text_at(nint responses,
int index)
```

Parameters

`responses` nint

The responses object.

`index` int

The index of the response.

Returns

nint

The response text. The returned string is owned by the Responses object and is valid only for its lifetime. Returns nint.Zero if index is out of bounds.

litert_lm_responses_get_score_at(nint, int)

Returns the score at a given index.

```
public static extern float litert_lm_responses_get_score_at(nint responses, int index)
```

Parameters

`responses` nint

The responses object.

`index` int

The index of the response.

Returns

float

The score. Returns 0.0f if index is out of bounds or no score is present.

litert_lm_responses_get_token_length_at(nint, int)

Returns the token length at a given index.

```
public static extern int litert_lm_responses_get_token_length_at(nint responses, int index)
```

Parameters

`responses` nint

The responses object.

`index` int

The index of the response.

Returns

int

The token length. Returns 0 if index is out of bounds or no token length is present.

litert_lm_responses_get_token_scores_at(nint, int)

Returns the token scores at a given index.

```
public static extern nint litert_lm_responses_get_token_scores_at(nint responses, int index)
```

Parameters

`responses` nint

The responses object.

`index` int

The index of the response.

Returns

nint

A pointer to the internal array of token scores. Returns nint.Zero if index is out of bounds or no token scores are present.

litert_lm_responses_has_score_at(nint, int)

Returns whether the response contains a score at the given index.

```
public static extern bool litert_lm_responses_has_score_at(nint responses, int index)
```

Parameters

`responses` nint

The responses object.

`index` int

The index of the response.

Returns

bool

[true](#) if the score is available at the given index, [false](#) otherwise.

litert_lm_responses_has_token_length_at(nint, int)

Returns whether the response contains a token length at the given index.

```
public static extern bool litert_lm_responses_has_token_length_at(nint responses, int index)
```

Parameters

`responses` nint

The responses object.

`index` int

The index of the response.

Returns

bool

[true](#) if the token length is available at the given index, [false](#) otherwise.

litert_lm_responses_has_token_scores_at(nint, int)

Returns whether the response contains token scores at the given index.

```
public static extern bool litert_lm_responses_has_token_scores_at(nint responses, int index)
```

Parameters

`responses` nint

The responses object.

`index` int

The index of the response.

Returns

bool

true if token scores are available at the given index, false otherwise.

Class NativeAPI.SamplerParams

Namespace: Uralstech.UAI.LiteRTLM.Native

```
public static class NativeAPI.SamplerParams
```

Inheritance

object ← NativeAPI.SamplerParams

Methods

litert_lm_sampler_params_create(SamplerType)

Creates LiteRT LM Sampler Parameters with a specific sampler type. The caller is responsible for destroying the parameters using litert_lm_sampler_params_delete(nint).

```
public static extern nint litert_lm_sampler_params_create(SamplerType type)
```

Parameters

`type` SamplerType

The sampler type to use.

Returns

nint

A pointer to the created parameters, or nint.Zero on failure.

litert_lm_sampler_params_delete(nint)

Destroys LiteRT LM Sampler Parameters.

```
public static extern void litert_lm_sampler_params_delete(nint @params)
```

Parameters

`params` nint

The parameters to destroy.

litert_lm_sampler_params_set_seed(nint, int)

Sets the seed.

```
public static extern void litert_lm_sampler_params_set_seed(nint @params, int seed)
```

Parameters

`params` nint

`seed` int

litert_lm_sampler_params_set_temperature(nint, float)

Sets the temperature.

```
public static extern void litert_lm_sampler_params_set_temperature(nint @params,
float temperature)
```

Parameters

`params` nint

`temperature` float

litert_lm_sampler_params_set_top_k(nint, int)

Sets the top-k value.

```
public static extern void litert_lm_sampler_params_set_top_k(nint @params, int topK)
```

Parameters

`params` nint

`topK` int

litert_lm_sampler_params_set_top_p(nint, float)

Sets the top-p value.

```
public static extern void litert_lm_sampler_params_set_top_p(nint @params, float topP)
```

Parameters

`params` nint

`topP` float

Class NativeAPI.Session

Namespace: Uralstech.UAI.LiteRTLM.Native

```
public static class NativeAPI.Session
```

Inheritance

object ← NativeAPI.Session

Methods

litert_lm_session_cancel_process(nint)

Cancels the current processing in the session.

```
public static extern void litert_lm_session_cancel_process(nint session)
```

Parameters

`session` nint

The session to cancel processing on.

litert_lm_session_delete(nint)

Destroys a LiteRT LM Session.

```
public static extern void litert_lm_session_delete(nint session)
```

Parameters

`session` nint

The session to destroy.

litert_lm_session_generate_content(nint, nint[], nuint)

Generates content from the input prompt.

```
public static extern nint litert_lm_session_generate_content(nint session, nint[] inputs,
nuint numInputs)
```

Parameters

`session` nint

The session to use for generation.

`inputs` nint[]

An array of InputData structs representing the multimodal input.

`numInputs` nuint

The number of InputData structs in the array.

Returns

nint

A pointer to the responses, or nint.Zero on failure. The caller is responsible for deleting the responses using [litert_lm_responses_delete(nint)](#).

litert_lm_session_generate_content_stream(nint, nint[], nuint, nint, nint)

Generates content from the input prompt and streams the response via a callback. This is a non-blocking call that will invoke the callback from a background thread for each chunk.

```
public static extern int litert_lm_session_generate_content_stream(nint session, nint[]
inputs, nuint numInputs, nint callback, nint callbackData)
```

Parameters

`session` nint

The session to use for generation.

`inputs` nint[]

An array of InputData structs representing the multimodal input.

`numInputs` nuint

The number of InputData structs in the array.

`callback` nint

The callback function (NativeAPI.StreamCallback) to receive response chunks.

`callbackData` nint

A pointer to user data that will be passed to the callback.

Returns

int

0 on success, non-zero on failure to start the stream.

litert_lm_session_get_benchmark_info(nint)

Retrieves the benchmark information from the session. The caller is responsible for destroying the benchmark info using litert_lm_benchmark_info_delete(nint).

```
public static extern nint litert_lm_session_get_benchmark_info(nint session)
```

Parameters

`session` nint

The session to get the benchmark info from.

Returns

nint

A pointer to the benchmark info, or nint.Zero on failure.

litert_lm_session_run_decode(nint)

Starts the decoding process for the model to predict the response based on the input prompt/query added after using [litert_lm_session_run_prefill(nint, nint[], nuint)](). This is a blocking call and the function will return when the decoding process is done.

```
public static extern nint litert_lm_session_run_decode(nint session)
```

Parameters

`session` nint

The session to use.

Returns

nint

A pointer to the responses, or nint.Zero on failure. The caller is responsible for deleting the responses using [litert_lm_responses_delete(nint)]().

litert_lm_session_run_decode_async(nint, nint, nint)

Starts the decoding process for the model to predict the response based on the input prompt/query added after using [litert_lm_session_run_prefill(nint, nint[], nuint)](). This is a non-blocking call that will stream responses via a callback.

```
public static extern int litert_lm_session_run_decode_async(nint session, nint callback,
nint callbackData)
```

Parameters

`session` nint

The session to use.

`callback` nint

The callback function ([NativeAPI.StreamCallback]()) to receive response chunks.

`callbackData` nint

A pointer to user data that will be passed to the callback.

Returns

int

0 on success, non-zero on failure.

litert_lm_session_run_prefill(nint, nint[], nuint)

Adds the input prompt/query to the model for starting the prefilling process. This is a blocking call and the function will return when the prefill process is done.

```
public static extern int litert_lm_session_run_prefill(nint session, nint[] inputs,
nuint numInputs)
```

Parameters

`session` nint

The session to use.

`inputs` nint[]

An array of InputData structs representing the multimodal input.

`numInputs` nuint

The number of InputData structs in the array.

Returns

int

0 on success, non-zero on failure.

litert_lm_session_run_text_scoring(nint, string[], nuint, bool)

Scores the target text after the prefill process is done.

```
public static extern nint litert_lm_session_run_text_scoring(nint session, string[]
targetText, nuint numTargets, bool storeTokenLengths)
```

Parameters

`session` nint

The session to use.

`targetText` string[]

An array of target text strings to score.

`numTargets` nuint

The number of strings in the targetText array.

`storeTokenLengths` bool

Whether to store the token lengths of the target texts in the responses.

Returns

nint

A pointer to the responses, or nint.Zero on failure. The caller is responsible for deleting the responses using litert_lm_responses_delete(nint).

Class NativeAPI.SessionConfig

Namespace: Uralstech.UAI.LiteRTLM.Native

```
public static class NativeAPI.SessionConfig
```

Inheritance

object ← NativeAPI.SessionConfig

Methods

litert_lm_session_config_create()

Creates a LiteRT LM Session Config. The caller is responsible for destroying the config using litert_lm_session_config_delete(nint).

```
public static extern nint litert_lm_session_config_create()
```

Returns

nint

A pointer to the created config, or nint.Zero on failure.

litert_lm_session_config_delete(nint)

Destroys a LiteRT LM Session Config.

```
public static extern void litert_lm_session_config_delete(nint config)
```

Parameters

`config` nint

The config to destroy.

litert_lm_session_config_set_apply_prompt_template(nint, bool)

Sets whether to apply prompt template for this session.

```
public static extern void litert_lm_session_config_set_apply_prompt_template(nint config,
bool applyPromptTemplate)
```

Parameters

`config` nint

The config to modify.

`applyPromptTemplate` bool

Whether to apply prompt template.

litert_lm_session_config_set_audio_lora_path(nint, string)

Sets the path to the Audio LoRA weights file.

```
public static extern int litert_lm_session_config_set_audio_lora_path(nint config,
string audioLoraPath)
```

Parameters

`config` nint

The config to modify.

`audioLoraPath` string

The path to the audio LoRA weights file.

Returns

int

0 on success, non-zero on failure.

litert_lm_session_config_set_lora_path(nint, string)

Sets the path to the LoRA weights file.

```
public static extern int litert_lm_session_config_set_lora_path(nint config,
string loraPath)
```

Parameters

`config` nint

The config to modify.

`loraPath` string

The path to the text LoRA weights file.

Returns

int

0 on success, non-zero on failure.

litert_lm_session_config_set_max_output_tokens(nint, int)

Sets the maximum number of output tokens per decode step for this session.

```
public static extern void litert_lm_session_config_set_max_output_tokens(nint config,
int maxOutputTokens)
```

Parameters

`config` nint

The config to modify.

`maxOutputTokens` int

The maximum number of output tokens.

litert_lm_session_config_set_sampler_params(nint, nint)

Sets the sampler parameters for this session config.

```
public static extern void litert_lm_session_config_set_sampler_params(nint config,
nint samplerParams)
```

Parameters

`config` nint

The config to modify.

`samplerParams` nint

The sampler parameters to use.

Delegate NativeAPI.StreamCallback

Namespace: Uralstech.UAI.LiteRTLM.Native

Callback for streaming responses.

```
public delegate void NativeAPI.StreamCallback(nint callbackData, nint chunk)
```

Parameters

`callbackData` nint

A pointer to user-defined data passed to the stream function.

`chunk` nint

A pointer to the stream chunk object. It's only valid for the duration of the call.

Class NativeAPI.StreamChunk

Namespace: Uralstech.UAI.LiteRTLM.Native

```
public static class NativeAPI.StreamChunk
```

Inheritance

object ← NativeAPI.StreamChunk

Methods

litert_lm_stream_chunk_get_error(nint)

Gets the error message associated with this chunk, if any.

```
public static extern nint litert_lm_stream_chunk_get_error(nint chunk)
```

Parameters

`chunk` nint

Returns

nint

Remarks

Returns nint.Zero if there is no error.

litert_lm_stream_chunk_get_text(nint)

Gets the text content of the chunk.

```
public static extern nint litert_lm_stream_chunk_get_text(nint chunk)
```

Parameters

`chunk` nint

Returns

nint

Remarks

The returned string is owned by the chunk and is only valid as long as the chunk is valid. Returns nint. Zero if there is no text content in this chunk (e.g. if it is an error or metadata-only chunk).

litert_lm_stream_chunk_is_final(nint)

Returns [true](#) if this is the final chunk of the stream.

```
public static extern bool litert_lm_stream_chunk_is_final(nint chunk)
```

Parameters

`chunk` nint

Returns

bool

Class NativeAPI.ThinkingConfig

Namespace: Uralstech.UAI.LiteRTLM.Native

```
public static class NativeAPI.ThinkingConfig
```

Inheritance

object ← NativeAPI.ThinkingConfig

Methods

litert_lm_thinking_config_create()

Creates a default LiteRT LM Thinking Config (enabled with infinite budget -1). The caller is responsible for destroying the config using litert_lm_thinking_config_delete(nint).

```
public static extern nint litert_lm_thinking_config_create()
```

Returns

nint

A pointer to the created config, or nint.Zero on failure.

litert_lm_thinking_config_delete(nint)

Destroys a LiteRT LM Thinking Config.

```
public static extern void litert_lm_thinking_config_delete(nint config)
```

Parameters

`config` nint

The config to destroy.

litert_lm_thinking_config_set_enable_thinking(nint, bool)

Sets whether thinking/reasoning generation is enabled.

```
public static extern void litert_lm_thinking_config_set_enable_thinking(nint config,
bool enableThinking)
```

Parameters

`config` nint

The config to modify.

`enableThinking` bool

Whether thinking is enabled.

litert_lm_thinking_config_set_thinking_token_budget(nint, int)

Sets the thinking token budget.

```
public static extern void litert_lm_thinking_config_set_thinking_token_budget(nint config,
int thinkingTokenBudget)
```

Parameters

`config` nint

The config to modify.

`thinkingTokenBudget` int

Budget for token-by-token reasoning generation (-1 for infinite).

Class NativeAPI.TokenUnion

Namespace: Uralstech.UAI.LiteRTLM.Native

```
public static class NativeAPI.TokenUnion
```

Inheritance

object ← NativeAPI.TokenUnion

Methods

litert_lm_token_union_delete(nint)

Destroys a LiteRT LM Token Union.

```
public static extern void litert_lm_token_union_delete(nint tokenUnion)
```

Parameters

`tokenUnion` nint

The token union to destroy.

litert_lm_token_union_get_ids(nint, out nint, out nuint)

Returns the token ids from a token union.

```
public static extern int litert_lm_token_union_get_ids(nint tokenUnion, out nint outTokens,
out nuint outNumTokens)
```

Parameters

`tokenUnion` nint

The token union.

`outTokens` nint

A pointer to receive the internal array of token ids. The received pointer is valid only for the lifetime of the TokenUnion object.

`outNumTokens` nuint

A pointer to receive the number of token ids.

Returns

int

0 on success, non-zero if the type is not Ids.

litert_lm_token_union_get_string(nint)

Returns the string value from a token union.

```
public static extern nint litert_lm_token_union_get_string(nint tokenUnion)
```

Parameters

`tokenUnion` nint

The token union.

Returns

nint

The string value, or nint.Zero if the type is not String. The returned string is owned by the TokenUnion object and is valid only for its lifetime.

litert_lm_token_union_get_type(nint)

Returns the type of the token union.

```
public static extern TokenUnionType litert_lm_token_union_get_type(nint tokenUnion)
```

Parameters

`tokenUnion` nint

The token union.

Returns

TokenUnionType

The type of the token union.

Class NativeAPI.TokenUnions

Namespace: Uralstech.UAI.LiteRTLM.Native

```
public static class NativeAPI.TokenUnions
```

Inheritance

object ← NativeAPI.TokenUnions

Methods

litert_lm_token_unions_delete(nint)

Destroys a LiteRT LM Token Unions object.

```
public static extern void litert_lm_token_unions_delete(nint tokens)
```

Parameters

`tokens` nint

The token unions object to destroy.

litert_lm_token_unions_get_num_tokens(nint)

Returns the number of token unions in the collection.

```
public static extern nuint litert_lm_token_unions_get_num_tokens(nint tokens)
```

Parameters

`tokens` nint

The token unions object.

Returns

nuint

The number of token unions.

litert_lm_token_unions_get_token_at(nint, nuint)

Returns the token union at a given index from a collection.

```
public static extern nint litert_lm_token_unions_get_token_at(nint tokens, nuint index)
```

Parameters

`tokens` nint

The token unions collection.

`index` nuint

The index of the token union.

Returns

nint

A pointer to the token union at the given index, or nint.Zero if the index is out of bounds. The caller is responsible for deleting the result using litert_lm_token_union_delete(nint).

Class NativeAPI.TokenizeResult

Namespace: Uralstech.UAI.LiteRTLM.Native

```
public static class NativeAPI.TokenizeResult
```

Inheritance

object ← NativeAPI.TokenizeResult

Methods

litert_lm_tokenize_result_delete(nint)

Destroys a LiteRT LM Tokenize Result.

```
public static extern void litert_lm_tokenize_result_delete(nint result)
```

Parameters

`result` nint

The tokenize result to destroy.

litert_lm_tokenize_result_get_num_tokens(nint)

Returns the number of token ids from a tokenize result.

```
public static extern nuint litert_lm_tokenize_result_get_num_tokens(nint result)
```

Parameters

`result` nint

The tokenize result.

Returns

nuint

The number of token ids.

litert_lm_tokenize_result_get_tokens(nint)

Returns the token ids from a tokenize result.

```
public static extern nint litert_lm_tokenize_result_get_tokens(nint result)
```

Parameters

`result` nint

The tokenize result.

Returns

nint

A pointer to the internal array of token ids. The returned pointer is valid only for the lifetime of the `result` object.